King Tao Jason **Ng**

Profile Summary

Senior Data Scientist and Analytics Leader with 10+ years' experience delivering analytics for regulatory and market surveillance solutions. Proven track record of leading cross-functional teams, deploying NLP and machine learning systems at scale, and influencing executive stakeholders. Specialises in translating complex regulatory requirements into production-grade data applications that enhance risk detection and operational efficiency.

Contact details

✉ jason.kingtao.ng@gmail.com
jasonkng
kingtaojasonng
kingtaojasonng

Personal information

Citizenship: **Australian**
Languages: **English**, **Chinese**

Experience

◇ Duties ★ Key Achievements

Senior Data Scientist **Jul 2022 - Present**
Data Analytics and AI, ASIC

◇ Led end-to-end delivery of ML and NLP initiatives, including the Licensing ML Proof of Concept, Knowledge Management Metadata Extraction, and Lessons Learned Lite Review.

★ Delivered Lessons Learned Lite Review and nominated for the *Australia Day Achievement Award*.

Senior Data Scientist (Secondment) **Sep 2025 - Feb 2026**
Superannuation & Life Insurance, ASIC

◇ Designed and implemented a red-flag detection system for superannuation funds, enhancing regulatory risk identification and informing supervisory actions.

Sessional Teaching Academic **Feb 2022 - Jun 2022**
School of Computing, Macquarie University

◇ Led and tutored a coursework for Document Processing and Semantic Technologies (COMP3220), guiding students through complex concepts and practical applications.

Senior Analytics Platform Owner **Jul 2021 - Jul 2022**
Chief Data and Analytics Office, ASIC

◇ Led enterprise adoption of Databricks, RStudio, and Python Servers, and the Data Science Lab, improving analytics productivity and standardising workflows across data teams.

Data Scientist **Oct 2019 - Jul 2021**
Chief Data and Analytics Office, ASIC

◇ Developed NLP Prospectuses, which reduces document review time by significant amount and provides actionable insights to the Corporation team.

Data Scientist **Aug 2017 - Jun 2019**
School of Computer Science, University of Sydney

◇ Designed and developed Researcher Collaboration Recommendation System using Python and D3.js, enabling data-driven networking through interactive visualizations.

★ Invited to present at the *16th Australasian Data Mining Conference*.

Analyst **Jul 2016 - Oct 2019**
Market Infrastructure, ASIC

◇ Proposed and implemented Wash Trade Detection analytics using KDB+/Q, strengthening market surveillance capabilities and improving identification of suspicious trading behaviour.

★ Successfully identified a confirmed wash trading case using Wash Trade Detection analytics

Market Technology Analyst **Jan 2013 - Jul 2016**
Market & Participant Supervision, ASIC

◇ Led the development and launch of the *Market Analysis and Intelligence Surveillance System*, enhancing market integrity.

★ Earned the *Australia Day Achievement Award* for outstanding accomplishment.

Core Competencies

- Advanced Analytics & Market Surveillance
- Machine Learning & NLP
- Data Governance
- Data Platform Strategy
- Stakeholder & Executive Engagement

Awards

- **Australia Day Achievement Award** — ASIC
- **ARI Award** — Aristocrat Technologies Australia
- **Competent Communicator** — Toastmasters International

Technical Stack

- Python, R, C/C++, KDB+/Q
- Databricks, Elastic Stack, Power BI
- Bayesian Methods, Statistics

Intranet Programmer **Oct 2011 - Dec 2011**
Information and Communications Technology, Scots College

◇ Oversaw the maintenance and management of IBM DB2, ensuring database reliability, performance, and security.

Domain Analyst **Mar 2011 - May 2011**
Wagering Technology, Tabcorp Holdings Limited

◇ Led and managed requirements gathering for the Customer Loyalty project, ensuring alignment with business objectives and stakeholder needs.

Software Engineer **Jul 2008 - Mar 2011**
Studio Development, Aristocrat Technologies Australia

◇ Designed and developed gaming applications using C and C++, creating engaging user experiences.

★ Successfully developed *Peeking Duck*, which received highly positive feedback from gaming floors.

Graduate Software Engineer **Jul 2007 - Jul 2008**
Platform Development, Aristocrat Technologies Australia

◇ Designed and developed networking protocols for casino management systems using C and C++, ensuring secure and efficient communication across the platforms.

★ Successfully proposed a third-party licensing initiative to the executive team, leading to the development of *JAW* gaming the following year.

Graduate Test Engineer **Jan 2007 - Jul 2007**
Software Integrity Testing, Aristocrat Technologies Australia

◇ Developed comprehensive test plans and performed unit and integration testing to ensure high-quality, reliable software delivery.

Education

Master of Research, *Macquarie University* **Jul 2021 - Oct 2023**

★ Identifying Causal Directions from Text: Unsupervised Learning using Bayesian Framework

Master of Data Science, *University of Sydney* **Mar 2016 - Jul 2019**

★ Academic Capability Mapping

Master of Economics, *University of Sydney* **Mar 2008 - Jul 2012**

Master of Information Technology, *UNSW* **Jul 2005 - Dec 2007**

Bachelor of Science, *UNSW* **Jul 2001 - Mar 2005**

Publications

Exploring Causal Directions through Word Occurrences: Semi-supervised Bayesian Classification Framework, ***King Tao Jason Ng*** *and Diego Mollá. The 21st Annual Workshop of the Australasian Language Technology Association, Melbourne, VIC, Australia, November 29 - December 1, 2023*

Hierarchical Word Mover Distance for Collaboration Recommender System, *Chao Sun,* ***King Tao Jason Ng****, Philip Henville and Roman Marchant. The 16th Australasian Data Mining Conference, AusDM 2018, Bathurst, NSW, Australia, November 28 - 30, 2018, Revised Selected Papers*